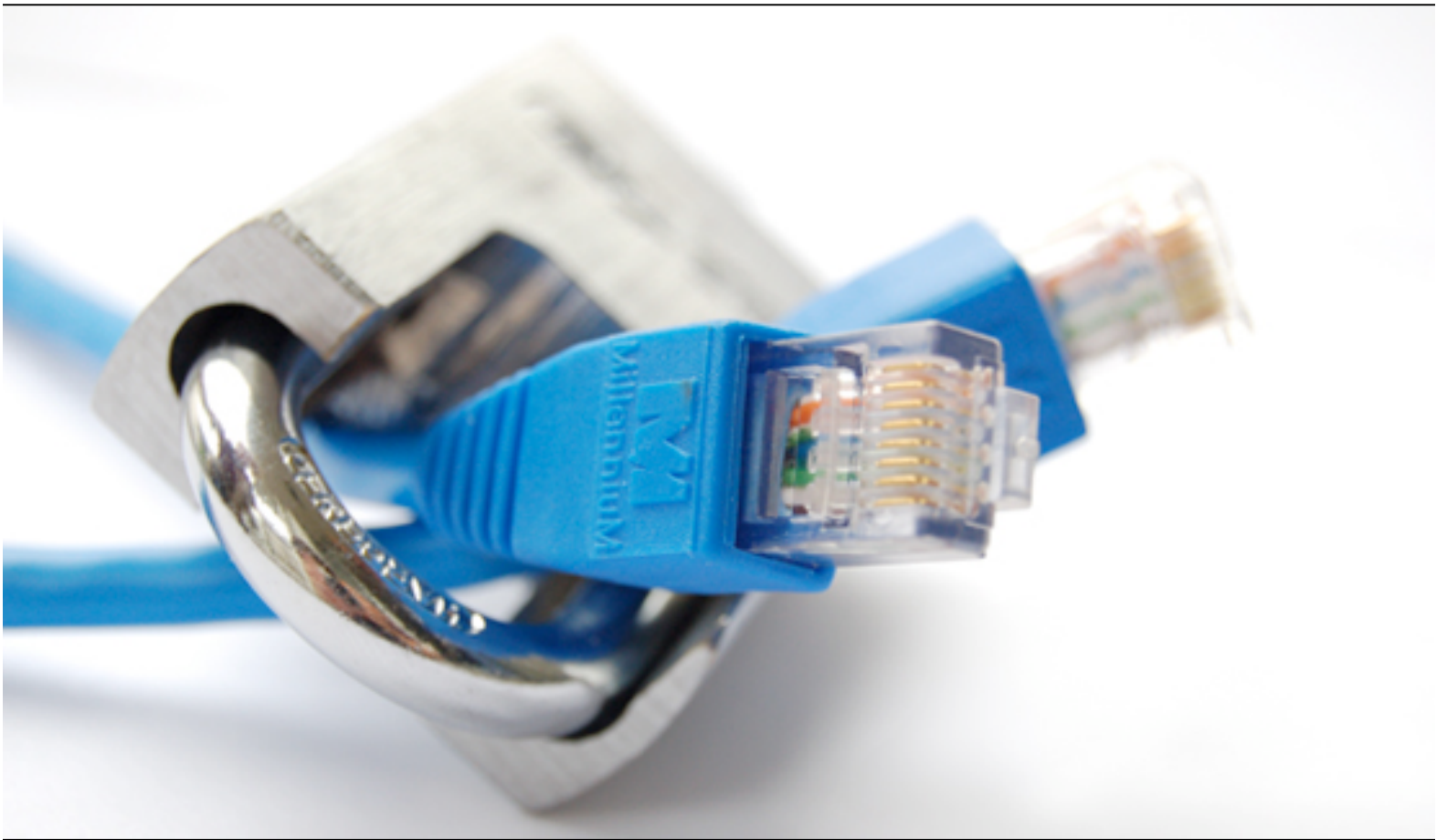


Atlantic Computational Excellence Network
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Security Policies

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Table of Contents

General	2
Resources and Facilities Covered by this Policy.....	2
Physical Security	2
Data Center Access	2
System Security	3
Network Security	3
Client-side Security	3
ACEnet Conditions of Use.....	3
Condition of Use Violation.....	4
Password Policy.....	4
Sharing and Data Security	4

General

We recognize that the information stored and generated on ACEnet resources is a vital asset to our researchers. The purpose of this Security Policy is to protect this asset and the proper functioning and availability of our resources by establishing responsibility for the security of the facility. This policy applies to all ACEnet users, staff and third parties involved in supporting the consortium.

Resources and Facilities Covered by this Policy

All resources managed by ACEnet are covered by this policy, including computing hardware and software, documentation and other reference materials, all data residing on ACEnet machines and all consortium owned data wherever it resides, media such as CD-ROM, tape and other storage devices, all researcher-owned hardware managed by ACEnet, and all other possessions managed by ACEnet.

Physical Security

Depending on the host institution, ACEnet houses its computational resources either in a self-managed data center or, when possible, co-housed with the institution's IT Services' data center. In either case, we expect at least the level of security outlined here.

Data Center Access

Data center access is limited to employees of ACEnet and, in co-housing arrangements, personnel as approved by local ITS policy. Keys are to be issued only to ACEnet employees whose duties extend to hands on hardware.

Visitors to our data centers must be accompanied at all times by a key-holding ACEnet employee. Approved contractors may perform duties in the data center unaccompanied however they are to rely on an ACEnet employee for access. In cases where ACEnet resources are co-housed, local access policies may be stricter, in which case we follow the local policy.

System Security

Network Security

ACEnet computational and storage resources must be behind a firewall with external access available only for approved services.

Privileged access must be performed over secure channels, such as encrypted network connections using SSH or IPSec.

Network-based vulnerability, compromise assessment and guideline compliance scans of systems and devices on ACEnet networks will be performed in order to detect known vulnerabilities, compromised hosts, and guideline compliance failures. These scans will be performed on all publicly addressed ACEnet servers at least every 6 months. Local ITS departments will be advised of this activity.

While ACEnet administration desires to provide a reasonable level of privacy, users should be aware that for security and network maintenance purposes, authorized individuals within ACEnet may monitor equipment, systems and network traffic at any time.

Client-side Security

ACEnet Conditions of Use

A user of ACEnet resources must agree to abide by the following conditions:

- A user is responsible for all activity performed using that account. Accounts are not to be shared and passwords must be kept private and secure.
- Accounts and allocations are to be used exclusively for the purposes and project(s) for which they have been approved.
- ACEnet resources are provided for research and may not be used for instructional or administrative purposes.
- It is a user's responsibility to ensure a valid license is held by the user or by ACEnet prior to installing or using commercial software.
- Users must respect the privacy of other users of the resources. ACEnet will routinely collect system and usage statistics for reporting purposes, but will abide by our privacy policy at all times.
- While ACEnet does have a user data protection strategy, it is a user's responsibility to ensure that they abide by the data protection and retention policies of their Universities and funders, independent of ACEnet.

Condition of Use Violation

Any user found to have violated this policy may be subject to disciplinary action including termination of all running jobs and account suspension. Except in emergency situations, no action will be taken without the consent of the Executive Director.

Password Policy

All passwords must meet the following requirements

- Contain both upper and lower case characters (e.g., a-z, A-Z)
- Have digits and punctuation characters as well as letters e.g., 0-9
#\$%^&*()_+|~-=\`{}[]:~<>?,./)
- Be at least 9 alphanumeric characters long.
- Are not a word in any language, slang, dialect, jargon, etc.
- Passwords should never be written down in an unsecured location or stored on-line.

User passwords will expire upon first login for new accounts (to change the initial password set by the technical staff) and then every year thereafter.

Sharing and Data Security

When accounts are created, the default Unix directory permissions are set to 0700, granting full control to the user, and blocking all others. Within those directories, the default umask is 022 which means files are created with world and group read permission (but, by default, can't be accessed except by the user due to the directory permissions.) Users may change permissions at their own discretion.

While ACEnet does have a user data protection strategy, it is a user's responsibility to ensure that they abide by the data protection and retention policies of their Universities and funders, independent of ACEnet.

The storage implemented by ACEnet is Sun's SAM-QFS, a hierarchical storage system using RAID 5 technology. The objective of a hierarchical storage system is large storage volume at low cost, while not sacrificing speed of access. The objectives of the RAID 5 technology are speed and fault tolerance.

Although there are features available in SAM-QFS that allow for backup and recovery of user data, SAM-QFS is not configured as a backup system at ACEnet. ACEnet endeavours to protect users from the effects of hardware failures, but we do not protect from accidental overwriting or deletion.

Given that ACEnet does not provide backup services, rather we "Archive" data, therefore users are strongly encouraged to make off-site (or multi-site) copies of their critical data, and to observe their home institution's data storage policies.

All ACEnet staff will respect the privacy of user data and will seek permission from the user before modifying or deleting anything in the user's directories. In the event of an emergency where deleting user files is necessary to maintain the security and functioning of the cluster, and

ACEnet Security Policies

attempts to contact the user his/her and project leader have failed, minimal file deletion will proceed only with permission of the Executive Director, and the user will be notified of the action.